

B. The probability density function (PDF)  $f$  of the random variable  $X$  is given by  $f(x) = \begin{cases} k; & a < x < b \\ 0; & \text{elsewhere} \end{cases}$  where  $k$  is a constant.

- State the type of the given probability distribution.
- Find the value(s) of  $k$ .
- Is it possible for a PDF for a continuous random variable to assign a value greater than one? Justify your answer.
- Prove that the standard deviation of the given probability distribution is given by  $\sigma = \frac{b-a}{\sqrt{12}}$ .
- The cumulative distribution function (CDF)  $F$  of a continuous random variable  $X$  is defined as  $F(x) = \int_{-\infty}^x f(x) dx$ , with  $f$  being the PDF.

For the PDF  $f$  highlighted in problem [B], find and sketch the CDF  $F$  as a function of  $x$  and find the value of  $F\left(\frac{3b+a}{4}\right)$ .

**Solution:**



B. a) uniform distribution

$$\text{b) } \because \int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow \int_a^b k dx = 1 \Rightarrow k \int_a^b dx = 1 \Rightarrow k[x]_a^b = 1 \Rightarrow k = \frac{1}{b-a}$$

c) yes.

$$\text{If } a = 0 \text{ and } b = 0.5 \Rightarrow k = 2 > 1$$



d)

$$\mu = E[X] = \int_{-\infty}^{\infty} x f(x) dx = \int_a^b x k dx = \frac{1}{b-a} \left[ \frac{x^2}{2} \right]_a^b = \frac{1}{2} \frac{b^2 - a^2}{b-a} = \frac{b+a}{2}$$

$$(E[X])^2 = \left( \frac{b+a}{2} \right)^2 = \frac{a^2 + 2ab + b^2}{4}$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_a^b x^2 k dx = \frac{1}{b-a} \left[ \frac{x^3}{3} \right]_a^b = \frac{1}{3} \frac{b^3 - a^3}{b-a}$$

$$\Rightarrow E[X^2] = \frac{1}{3} \frac{(b-a)(b^2 + ab + a^2)}{b-a} = \frac{b^2 + ab + a^2}{3}$$

$$\sigma^2 = E[X^2] - (E[X])^2 = \frac{4b^2 + 4ab + 4a^2}{12} - \frac{3a^2 + 6ab + 3b^2}{12} = \frac{a^2 - 2ab + b^2}{12}$$

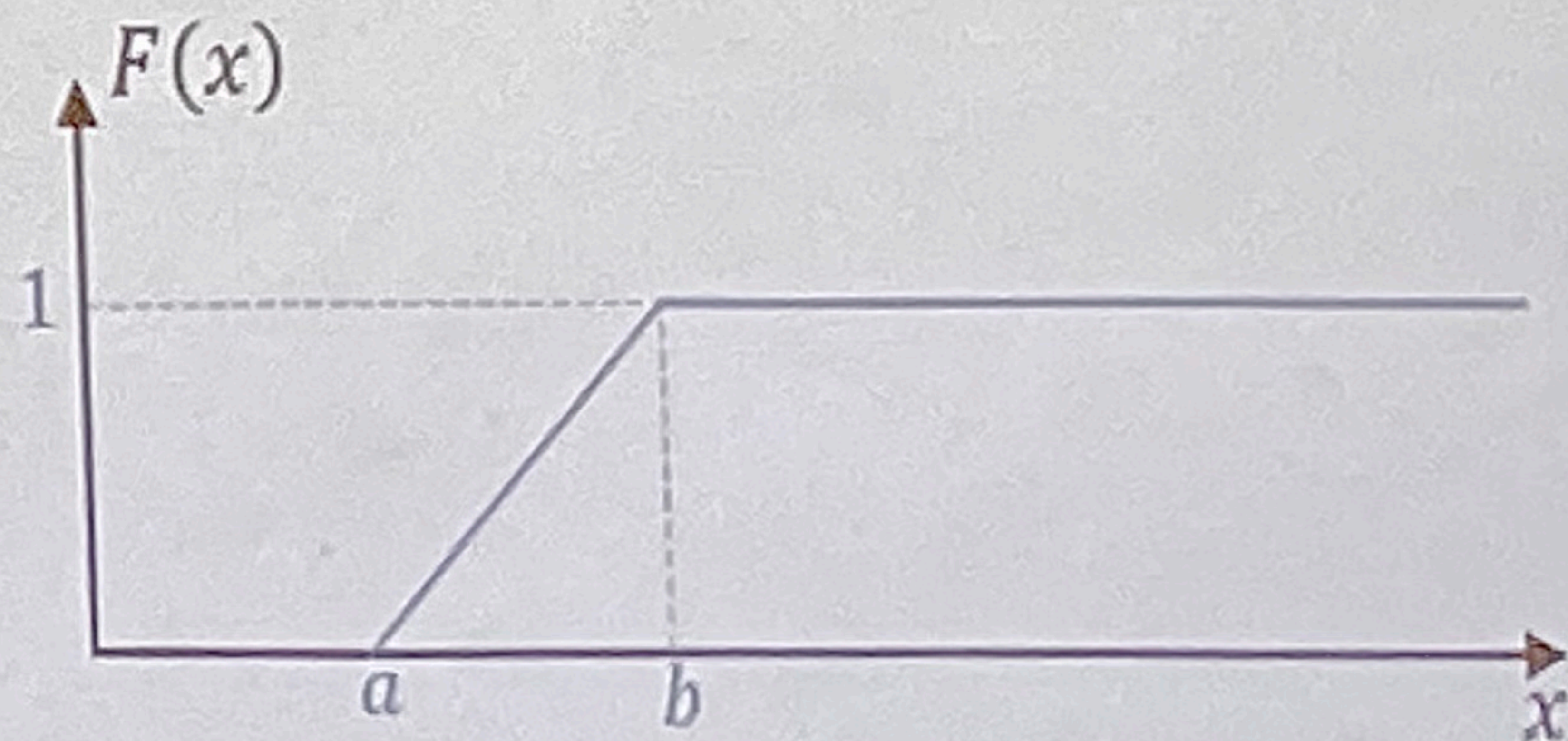
$$\Rightarrow \sigma^2 = \frac{(b-a)^2}{12} \Rightarrow \sigma = \frac{b-a}{\sqrt{12}}$$



$$e) F(x) = \int_{-\infty}^x f(t) dt$$

$$\text{For } a < x < b, F(x) = \int_a^x \frac{1}{b-a} dt = \left[ \frac{t}{b-a} \right]_a^x = \frac{1}{b-a} x - \frac{a}{b-a}$$

$$F(x) = \begin{cases} 0 & ; x \leq a \\ \frac{1}{b-a} x - \frac{a}{b-a} & ; a < x < b \\ 1 & ; x \geq b \end{cases}$$



$$F\left(\frac{3b+a}{4}\right) = \int_a^{\frac{3b+a}{4}} \frac{1}{b-a} dt = \frac{1}{b-a} \left( \frac{3b}{4} + \frac{a}{4} - \frac{4a}{4} \right) = \frac{1}{b-a} \frac{3b-3a}{4} = \frac{3(b-a)}{4(b-a)} = 0.75$$